

Author: Probably, you are right. However, the accepted notation is $\Delta f(x_0)$. Such a notation emphasized the fact that the increment of f (in other words, the given function of Δx) is referred to point x_0 . With the concepts of the increment introduced, it is not difficult to evaluate the rate of change of f close to x_0 .

Reader: This rate should be described by the ratio $\frac{\Delta f(x_0)}{\Delta x}$. For instance, if we compare $\Delta f(x_0)$ with an increment of f at another point Δx from the domain of the function (say, point $x = x_1$), we may obtain an inequality and therefore conclude that the rate of change of f close to point x_1 is greater than that close to x_0 .

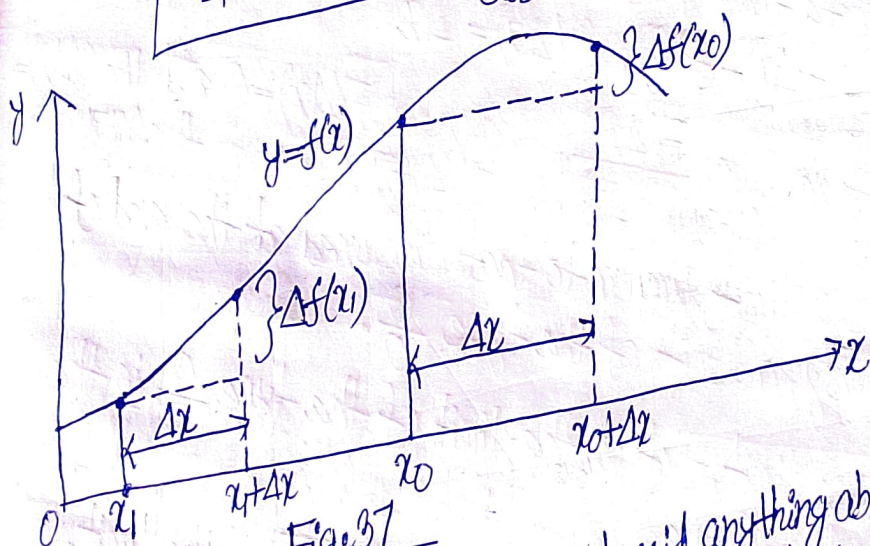


Fig. 37

Author: Please be careful. You have not said anything about the value of the increment Δx . If Δx is too large, the inequality you have just mentioned may lead to a wrong conclusion. I shall make myself clearer by referring to Fig. 37. As you see, $\frac{\Delta f(x_1)}{\Delta x} > \frac{\Delta f(x_0)}{\Delta x}$. You must agree, however, that close to point x_0 the function changes much faster (the graph of the function has a steeper slope) than in the vicinity of x_1 .

Reader: It is necessary that the value of the increment Δx be sufficiently small. The smaller is Δx the more accurate is the information about the rate of change of the function close to the point under consideration.

Author: Well, we can do even better than this. We may, for example, consider the limit of the ratio $\frac{\Delta f(x_0)}{\Delta x}$ for $\Delta x \rightarrow 0$ (remember the previous dialogue).